DESIGN DATA BOOK — SCREWED JOINTS & FASTENERS

Standard Engineering Design Data, Formulas & Solved Protocols (IS: 4218 - 1976)

Fastener Joint Classification & Design Index

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SECTION REF: Standard Design Reference Tables (IS: 4218 - 1976)

Table 11.1: Design Dimensions of Standard Metric Screw Threads

Thread	Pitch p	Nominal d	Pitch d_p	Core d_c	Area A_c
M 4	0.7 mm	4 mm	3.545 mm	3.141 mm	8.78 mm²
M 5	0.8 mm	5 mm	4.480 mm	4.019 mm	14.2 mm²
M 6	1.0 mm	6 mm	5.350 mm	4.773 mm	20.1 mm²
M 8	1.25 mm	8 mm	7.188 mm	6.466 mm	36.6 mm²
M 10	1.5 mm	10 mm	9.026 mm	8.160 mm	58.3 mm²
M 12	1.75 mm	12 mm	10.863 mm	9.858 mm	84.0 mm²
M 14	2.0 mm	14 mm	12.701 mm	11.546 mm	115 mm²
M 16	2.0 mm	16 mm	14.701 mm	13.546 mm	157 mm²
M 18	2.5 mm	18 mm	16.376 mm	14.933 mm	192 mm²
M 20	2.5 mm	20 mm	18.376 mm	16.933 mm	245 mm²

Table 11.2: Joint Factors (K)

Type of Joint / Gasket	Value of K
Metal to metal joint	0.00 to 0.10
Hard copper gasket	0.25 to 0.50
Soft copper gasket	0.50 to 0.75
Soft packing (through bolts)	0.75 to 1.00
Soft packing (studs)	1.00

Table 11.1 (Fine Series)

Fine Thread	Pitch p	Nominal d	Core d_c	Area A_c
M 8 x 1	1.0 mm	8 mm	6.773 mm	39.2 mm²
M 10 x 1.25	1.25 mm	10 mm	8.466 mm	61.6 mm²
M 12 x 1.25	1.25 mm	12 mm	10.466 mm	92.1 mm²

M 22	2.5 mm	22 mm	20.376 mm	18.933 mm	303 mm²
M 24	3.0 mm	24 mm	22.051 mm	20.320 mm	353 mm²
M 27	3.0 mm	27 mm	25.051 mm	23.320 mm	459 mm²
M 30	3.5 mm	30 mm	27.727 mm	25.706 mm	561 mm²
M 33	3.5 mm	33 mm	30.727 mm	28.706 mm	694 mm²
M 36	4.0 mm	36 mm	33.402 mm	31.093 mm	817 mm²
M 42	4.5 mm	42 mm	39.077 mm	36.416 mm	1104 mm²
M 48	5.0 mm	48 mm	44.752 mm	41.795 mm	1465 mm²
M 52	5.0 mm	52 mm	48.752 mm	45.795 mm	1755 mm²
M 56	5.5 mm	56 mm	52.428 mm	49.177 mm	2022 mm²

M 14 x 1.5	1.5 mm	14 mm	12.160 mm	125 mm²
M 16 x 1.5	1.5 mm	16 mm	14.160 mm	167 mm²
M 20 x 1.5	1.5 mm	20 mm	18.160 mm	272 mm²
M 24 x 2	2.0 mm	24 mm	21.546 mm	384 mm²

Empirical Core Diameter Approximation:

If table values are unavailable:

$d_c \approx 0.84d$

$A_c \approx \frac{\pi}{4}(0.84d)^2$

SECTION 1: Standard Tensile Capacity Analysis (M 30 Fasteners)

System Parameters	**Design Protocol**
Nominal Bolt Diameter: $d = 30\ \text{mm}$ (M 30 Coarse Series)	Retrieve core stress area A_c from Table 11.1
Safe Tensile Stress: $\sigma_t = 42\ \text{MPa} = 42\text{N/mm}^2$	Calculate safe tensile load capacity $P = A_c \sigma_t$
Preload $F_i = 0$	

1. Core Stress Area Lookup (A_c)

Table 11.1 lookup for M 30 coarse series thread

$$A_c = f(d)$$
$$= 561\text{mm}^2$$

2. Safe Tensile Load Capacity (P)

Maximum allowable axial tension force without initial tightening

$$P = A_c \times \sigma_t$$
$$= 561\text{mm}^2 \times 42\text{N/mm}^2$$
$$= 23562\ \text{N}$$
$$= \mathbf{23.562\ kN}$$

3. Design Output

Safe Tensile Load $P = \mathbf{23.562\ kN}$

SECTION 2: Initial Tightening Stress & Preload Analysis (Tap Bolts)

System Parameters	**Design Protocol**
Nominal Tap Bolt Size: $d = 24$ mm (M 24 Coarse Series)	Lookup core root diameter d_c from Table 11.1
Clamped Rigid Machine Joint	Calculate initial tightening load $P = 2840d$
Neglect external separation load	Calculate induced tensile stress $\sigma_t = \frac{P}{\frac{\pi}{4}d_c^2}$

1. Core Root Diameter Lookup (d_c)

Table 11.1 (coarse series) lookup for M 24 thread

$$d_c = 20.32 \text{ mm}$$

2. Initial Tightening Load (P)

Empirical initial tightening tension equation

$$P = 2840 \cdot d$$
$$= 2840 \times 24$$
$$= 68160 \text{ N}$$

3. Induced Tightening Stress (σ_t)

Initial tension force balance equation

$$68160 = \frac{\pi}{4}(d_c)^2 \cdot \sigma_t$$

$$= \frac{\pi}{4}(20.32)^2 \cdot \sigma_t$$

$$= 324 \cdot \sigma_t$$

$$\sigma_t = \frac{68160}{324}$$

$$= 210\mathrm{N/mm^2}$$

$$= \mathbf{210\ MPa}$$

4. Design Output

Tightening Stress $\boldsymbol{\sigma_t = 210\ \mathrm{MPa}}$

SECTION 3: Lifting Eye Bolt Sizing & Thread Selection

System Parameters	**Design Protocol**
Lifting Load: $P = 60\ \mathrm{kN} = 60000\ \mathrm{N}$	Equate lifting load to tensile capacity $P = \frac{\pi}{4} d_c^2 \sigma_t$
Permissible Tensile Stress: $\sigma_t = 100\ \mathrm{MPa} = 100 \mathrm{N/mm}^2$	Solve required core root diameter d_c
Thread Series: ISO Metric Coarse	Select standard bolt from Table 11.1

1. Required Core Root Diameter (d_c)

Axial tension load equation for lifting eye bolt

$$P = \frac{\pi}{4}(d_c)^2 \cdot \sigma_t$$

$$60000 = \frac{\pi}{4}(d_c)^2 \times 100$$

$$= 78.55(d_c)^2$$

$$(d_c)^2 = \frac{60000}{78.55}$$

$$= 763.84$$

$$d_c = \sqrt{763.84}$$

$$= 27.64\ \mathrm{mm}$$

2. Standard Thread Selection (Table 11.1)

Select standard coarse series thread with $d_c \geq 27.64\ \mathrm{mm}$

$$d_{\mathrm{c,\ std}} \geq d_c$$

$$d_c = 28.706\ \mathrm{mm}$$

$$d = 33\ \mathrm{mm}$$

3. Design Output

Select Fastener Designation: M 33 Bolt ($d = 33\ \mathrm{mm}, d_c = 28.706\ \mathrm{mm}$)

SECTION 3: Lifting Eye Bolt — MECHANICAL DIAGRAM

11.13 Stress due to Combined Forces

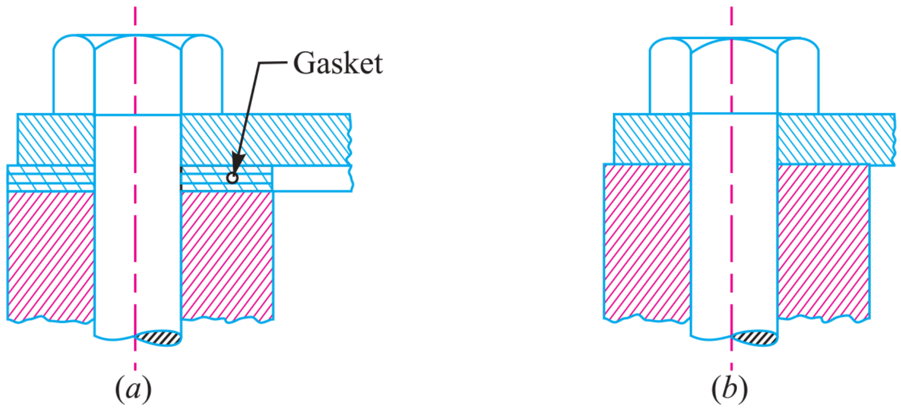


Fig. 11.23

Figure 11.23: Structural Lifting Eye Bolt Configuration and Load Path

SECTION 4: Flanged Shaft Coupling Fastener Sizing under Torsional Shear

System Parameters	**Design Protocol**
Transmitted Torque: $T = 25\ \mathrm{N \cdot m} = 25000\ \mathrm{N \cdot mm}$	Calculate total tangential shearing load $P_s = \frac{T}{R_p}$
Pitch Circle Radius: $R_p = 30\ \mathrm{mm}$	Equate P_s to shear capacity of n bolts $P_s = n\left(\frac{\pi}{4} d_c^2\right)\tau$
Bolt Count: $n = 4$ | Allowable Shear Stress: $\tau = 30\ \mathrm{MPa} = 30 \mathrm{N/mm^2}$	Select standard bolt size from Table 11.1

1. Total Shearing Load (P_s)

Total tangential shearing force acting at PCD

$$P_s = \frac{T}{R_p}$$

$$= \frac{25000}{30}$$

$$= 833.3\ \mathrm{N}$$

2. Required Core Diameter (d_c)

Solve root diameter for $n = 4$ bolts carrying shear

$$P_s = n \cdot \left[\frac{\pi}{4}(d_c)^2\right] \cdot \tau$$

$$833.3 = 4 \cdot \left[\frac{\pi}{4}(d_c)^2\right] \times 30$$

$$= 94.26(d_c)^2$$

$$(d_c)^2 = \frac{833.3}{94.26}$$

$$= 8.84$$

$$d_c = \sqrt{8.84}$$

$$= 2.97 \text{ mm}$$

3. Standard Thread Selection (Table 11.1)

Select standard coarse series thread with $d_c \geq 2.97 \text{ mm}$

$$d_{\text{c, std}} \geq d_c$$

$$d_c = 3.141 \text{ mm}$$

$$d = 4 \text{ mm}$$

4. Design Output

Select Fastener Designation: M 4 Bolt ($d = 4$ mm, $d_c = 3.141$ mm)

SECTION 5: Safety Valve Fulcrum Pin Sizing & Lever Equilibrium

System Parameters	Design Protocol
Valve Diameter: $D = 100\ \text{mm}$	Calculate upward steam force $F = \frac{\pi}{4}D^2p$
Blow-off Steam Pressure: $p = 1.6\text{N}/\text{mm}^2$	Apply lever moment equilibrium for end load W and fulcrum load $P = F - W$
Leverage Ratio: $\frac{L_1}{L_2} = 8$ | Permissible Tensile Stress: $\sigma_t = 50\ \text{MPa}$	Size fine metric thread d_c for fulcrum pin

1. Total Upward Force on Valve (F)

Total steam pressure load on valve disc

$$F = \frac{\pi}{4}D^2 \cdot p$$

$$= \frac{\pi}{4}(100)^2 \times 1.6$$

$$= 12568\ \text{N}$$

2. Lever End Load (W) & Fulcrum Load (P)

Static moment equilibrium for Type 1 lever

$$W = \frac{F}{\frac{L_1}{L_2}}$$

$$= \frac{12568}{8}$$

$$= 1571 \text{ N}$$

$$P = F - W$$

$$= 12568 - 1571$$

$$= 10997 \text{ N}$$

3. Required Core Diameter (d_c)

Solve root core diameter for fulcrum screw

$$P = \frac{\pi}{4}(d_c)^2 \cdot \sigma_t$$

$$10997 = \frac{\pi}{4}(d_c)^2 \times 50$$

$$= 39.27(d_c)^2$$

$$(d_c)^2 = \frac{10997}{39.27}$$

$$= 280$$

$$d_c = \sqrt{280}$$

$$= 16.7 \text{ mm}$$

4. Standard Fine Thread Selection (Table 11.1)

Select fine series metric thread with $d_c \geq 16.7$ mm

$$d_{\text{c, std}} \geq d_c$$

$$d_c = 18.376 \text{ mm}$$

Fine thread: M 20 x 1.5

5. Design Output

Select Fine Series Thread: M 20 x 1.5 ($d = 20$ mm, $d_c = 18.376$ mm)

SECTION 6: Cylinder Head Cover Stud Layout & Leak-Proof Pitch Verification

System Parameters	**Design Protocol**
Cylinder Effective Diameter: $D = 350 \text{ mm}$	Calculate total upward steam load $P = \frac{\pi}{4}D^2p$
Maximum Steam Pressure: $p = 1.25 \text{N/mm}^2$	Assume M 24 studs ($d_c = 20.32 \text{ mm}$) to find count
Permissible Stud Stress: $\sigma_t = 33 \text{ MPa} = 33 \text{N/mm}^2$	$n = \frac{P}{P_1}$
	Verify leak-proof pitch limits $20\sqrt{d_1} \le p_c \le 30\sqrt{d_1}$

1. Total Upward Steam Load (P)

Total gas pressure load on cylinder cover

$$P = \frac{\pi}{4}D^2 \cdot p$$

$$= \frac{\pi}{4}(350)^2 \times 1.25$$

$$= 120265 \text{ N}$$

2. Capacity per M 24 Stud (P_1) & Stud Count (n)

Resisting force capacity per M 24 stud ($d_c = 20.32$ mm)

$$P_1 = \frac{\pi}{4}(d_c)^2 \cdot \sigma_t$$

$$= \frac{\pi}{4}(20.32)^2 \times 33$$

$$= 10700 \text{ N}$$

$$n = \frac{P}{P_1}$$

$$= \frac{120265}{10700}$$

$$= 11.24 \Rightarrow \boldsymbol{n = 12 \text{ studs}}$$

3. Pitch Circle Diameter (D_p) & Circumferential Pitch (p_c)

Flange wall $t = 10$ mm, stud hole $d_1 = 25$ mm

$$D_p = D + 2t + 3d_1$$

$$= 350 + 2(10) + 3(25)$$

$$= 445 \text{ mm}$$

$$p_c = \frac{\pi D_p}{n}$$

$$= \frac{\pi \times 445}{12}$$

$$= 116.5 \text{ mm}$$

4. Leak-Proof Pitch Verification

Empirical leak-tight pitch limits (d_1 in mm)

$$p_c, \text{min} = 20\sqrt{d_1}$$

$$= 20\sqrt{25}$$

$$= 100 \text{ mm}$$

$$p_c, \text{max} = 30\sqrt{d_1}$$

$$= 30\sqrt{25}$$

$$= 150 \text{ mm}$$

$$100 \text{ mm} \leq 116.5 \text{ mm} \leq 150 \text{ mm} \ (\textbf{SAFE} \ \& \ \textbf{SATISFACTORY}$$

5. Design Output

Use 12 Studs of M 24 Size ($n = 12$, M 24 Studs)

SECTION 6: Cylinder Cover Stud Layout — MECHANICAL DIAGRAM

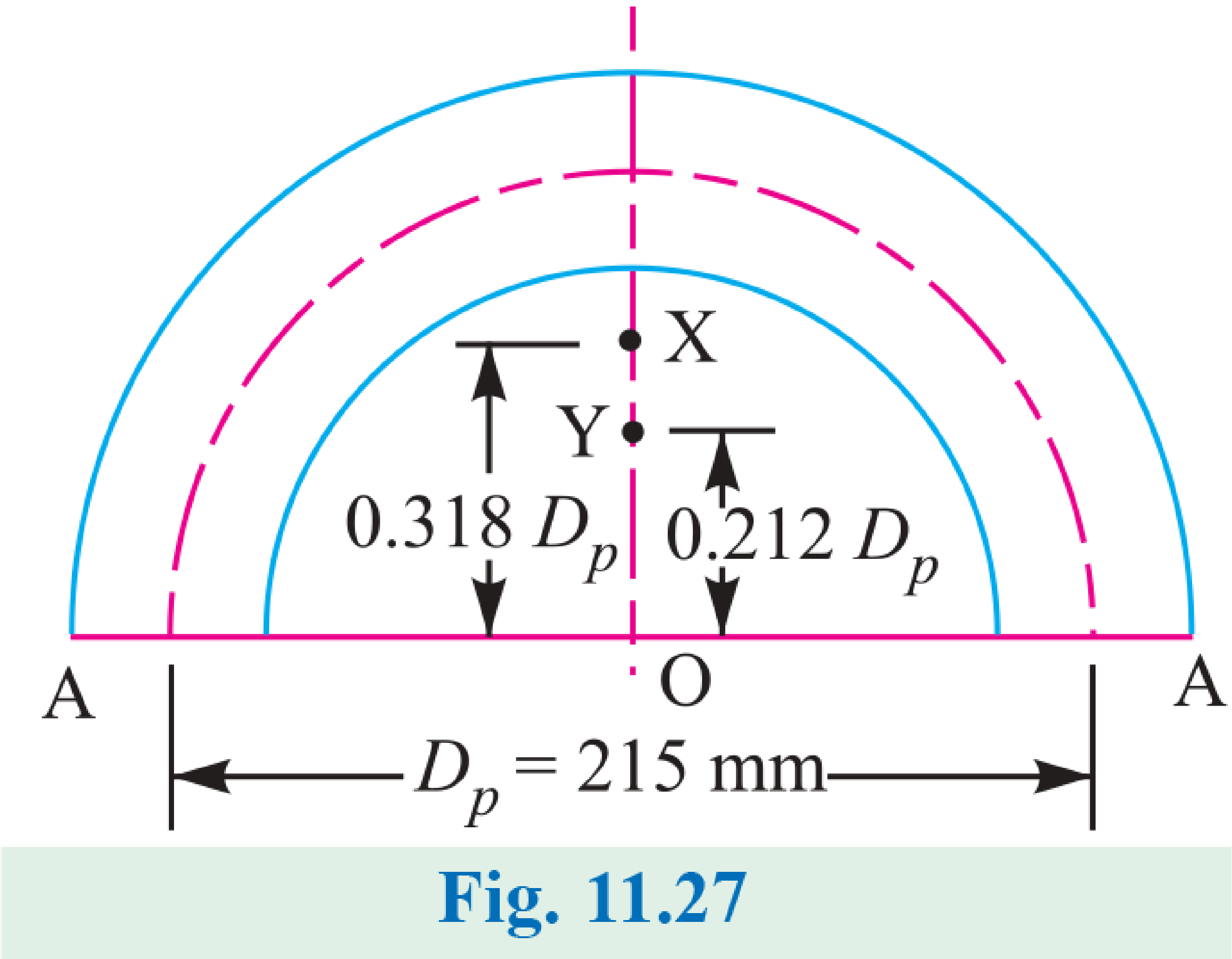


Figure 11.27: Cylinder Head Joint & Stud Pitch Circle Assembly

SECTION 7: Pressure Vessel Inspection Cover Plate & Bolt Sizing

System Parameters	**Design Protocol**
Inspection Hole Diameter: $D = 120\ \mathrm{mm}$ ($r = 60\ \mathrm{mm}$)	Part 1: Calculate shell wall thickness t via Lame
Internal Vessel Pressure: $p = 6\mathrm{N}/\mathrm{mm}^2$	Part 2: Determine bolt count n for M 24 and check pitch p_c
Plate Tensile Stress: $\sigma_t = 60\ \mathrm{MPa}$ | Bolt Stress: $\sigma_{\mathrm{tb}} = 40\ \mathrm{MPa}$	Part 3: Calculate cover plate thickness t_1 under bending moment

1. Shell Wall Thickness (t)

Lame equation for thick pressure vessels

$$t = r\left[\sqrt{\frac{\sigma_t + p}{\sigma_t - p}} - 1\right]$$

$$= 60\left[\sqrt{\frac{60+6}{60-6}} - 1\right]$$

$$= 60\left[\sqrt{1.222} - 1\right]$$

$$= 6.3\ \mathrm{mm}$$

Adopt Shell Thickness $t = 10$ mm

2. Fixing Bolts Sizing & Count (n)

Upward pressure load P and M 24 bolt capacity ($d_c = 20.32$ mm)

$$P = \frac{\pi}{4}D^2 \cdot p$$

$$= \frac{\pi}{4}(120)^2 \times 6$$

$$= 67860 \text{ N}$$

$$P_1 = \frac{\pi}{4}(d_c)^2 \cdot \sigma_{\text{tb}}$$

$$= \frac{\pi}{4}(20.32)^2 \times 40$$

$$= 12973 \text{ N}$$

$$n = \frac{P}{P_1}$$

$$= \frac{67860}{12973}$$

$$= 5.23 \Rightarrow \boldsymbol{n = 6 \text{ bolts}}$$

3. PCD & Leak-Proof Pitch Verification

Bolt hole $d_1 = 25$ mm for M 24 bolts

$$D_p = D + 2t + 3d_1$$

$$= 120 + 2(10) + 3(25)$$

$$= 215 \text{ mm}$$

$$p_c = \frac{\pi D_p}{n}$$

$$= \frac{\pi \times 215}{6}$$

$$= 112.6 \text{ mm}$$

$$100 \text{ mm} \leq 112.6 \text{ mm} \leq 150 \text{ mm} \ (\textbf{SAFE} \ \& \ \textbf{SATISFACTORY}$$

4. Cover Plate Thickness (t_1)

Bending moment $M = 0.053PD_p$ and section modulus $Z = \frac{1}{6}w(t_1)^2$

$$M = 0.053PD_p$$

$$= 0.053 \times 67860 \times 215$$

$$= 773265 \text{ N} \cdot \text{mm}$$

$$D_o = D_p + 3d_1$$

$$= 290 \text{ mm}$$

$$w = D_o - 2d_1$$

$$= 240 \text{ mm}$$

$$Z = \frac{1}{6} w (t_1)^2$$

$$= \frac{1}{6} \times 240 \times (t_1)^2$$

$$= 40 (t_1)^2$$

$$60 = \frac{773265}{40 (t_1)^2}$$

$$(t_1)^2 = 322.2$$

$$t_1 = 17.95 \text{ mm}$$

$$= \mathbf{18\ mm}$$

5. Design Output

Final Design: 6 Bolts of M 24 Size, Cover Plate Thickness $\boldsymbol{t_1 = 18}$ mm

SECTION 8: Preloaded Flange Fasteners with Gasket Compression (Soft Copper)

System Parameters	**Design Protocol**
Steam Pressure: $p = 0.7\text{N/mm}^2$	Calculate external load per bolt $P_2 = \frac{P_{\text{total}}}{12}$
Number of Bolts: $n = 12$	Formulate resultant axial load $P = 2840d + KP_2$
Effective Cylinder Diameter: $D = 300\ \text{mm}$	Solve governing quadratic for nominal diameter d
Allowable Stress: $\sigma_t = 100\ \text{MPa}$ | Soft Copper Gasket	
Ratio: $K = 0.5$	

1. External Pressure Load per Bolt (P_2)

Upward steam pressure load shared by 12 bolts

$$P_{\text{total}} = \frac{\pi}{4}D^2 \cdot p$$

$$= \frac{\pi}{4}(300)^2 \times 0.7$$

$$= 49480\ \text{N}$$

$$P_2 = \frac{P_{\text{total}}}{n} = \frac{49480}{12} = 4124 \text{ N}$$

2. Resultant Axial Load per Bolt (P)

Initial tension $P_1 = 2840d$ plus gasket compression load

$$P = P_1 + K \cdot P_2 = 2840d + 0.5 \times 4124 = 2840d + 2062 \text{ N}$$

3. Core Resisting Capacity & Quadratic Equation

Core diameter relation $d_c = 0.84d$

$$P = \frac{\pi}{4}(d_c)^2 \cdot \sigma_t = \frac{\pi}{4}(0.84d)^2 \times 100 = 55.4d^2$$

$$55.4d^2 = 2840d + 2062$$

$$55.4d^2 - 2840d - 2062 = 0$$

$$d^2 - 51.3d - 37.2 = 0$$

4. Quadratic Diameter Solution (d)

Quadratic formula evaluation for nominal bolt diameter

$$d = \frac{51.3 \pm \sqrt{(51.3)^2 - 4(1)(-37.2)}}{2}$$
$$= \frac{51.3 + \sqrt{2631.69 + 148.8}}{2}$$
$$= \frac{51.3 + 52.73}{2}$$
$$= \mathbf{52.01\ mm}$$

5. Design Output

Select Fastener Designation: M 52 Bolt ($d = 52$ mm)

SECTION 9: Soderberg Fatigue & Preload Design Criterion

System Parameters	**Design Protocol**
$D = 300 \text{ mm}, p = 1.5\text{N/mm}^2, n = 8, \sigma_y = 330 \text{ MPa}, \sigma_e = 240 \text{ MPa}$ Initial Preload: $P_1 = 1.5P_2$, Gasket Factor: $K = 0.5$, "F.S." = 2	Calculate maximum load $P_1 + KP_2$ & minimum load P_1 per bolt Determine mean load P_m and variable load P_v per bolt Apply Soderberg fatigue equation to solve core diameter d_c

1. Total Steam Load (P_2) & Initial Preload (P_1)

Steam pressure force and initial preload tension

$$P_2 = \frac{\pi}{4} D^2 \cdot p$$

$$= \frac{\pi}{4}(300)^2 \times 1.5$$

$$= 106040 \text{ N}$$

$$P_1 = 1.5P_2$$
$$= 1.5 \times 106040$$
$$= 159060 \text{ N}$$

2. Maximum & Minimum Load per Bolt

Load allocation shared across $n = 8$ bolts

$$P_{\max} = P_1 + KP_2$$
$$= 159060 + 0.5(106040)$$
$$= 212080 \text{ N}$$
$$P_{\text{max, bolt}} = \frac{212080}{8}$$
$$= 26510 \text{ N}$$
$$P_{\text{min, bolt}} = \frac{159060}{8}$$
$$= 19882 \text{ N}$$

3. Mean (P_m) & Variable (P_v) Loads per Bolt

Mean and alternating cyclic load components

$$P_m = \frac{P_{\text{max, bolt}} + P_{\text{min, bolt}}}{2}$$

$$= \frac{26510 + 19882}{2}$$

$$= 23196 \text{ N}$$

$$P_v = \frac{P_{\text{max, bolt}} - P_{\text{min, bolt}}}{2}$$

$$= \frac{26510 - 19882}{2}$$

$$= 3314 \text{ N}$$

4. Mean (σ_m) & Variable (σ_v) Stresses

Stress area $A_c = \frac{\pi}{4}(d_c)^2 = 0.7854(d_c)^2$

$$\sigma_m = \frac{P_m}{A_c}$$

$$= \frac{23196}{0.7854(d_c)^2}$$

$$= \frac{29534}{(d_c)^2} \text{N/mm}^2$$

$$\sigma_v = \frac{P_v}{A_c}$$

$$= \frac{3314}{0.7854(d_c)^2}$$

$$= \frac{4220}{(d_c)^2} \mathrm{N/mm^2}$$

5. Soderberg Fatigue Failure Equation

Soderberg line equation substitution

$$\sigma_v = \left(\frac{\sigma_e}{\text{F.S.}}\right)\left[1 - \frac{\sigma_m \cdot \text{F.S.}}{\sigma_y}\right]$$

$$\frac{4220}{(d_c)^2} = \left(\frac{240}{2}\right)\left[1 - \left(\frac{29534}{(d_c)^2}\right) \cdot \left(\frac{2}{330}\right)\right]$$

$$= 120\left[1 - \frac{179.0}{(d_c)^2}\right]$$

$$\frac{4220 + 21480}{(d_c)^2} = 120$$

$$\frac{25700}{(d_c)^2} = 120$$

$$d_c = \sqrt{214.17}$$

$$= 14.63 \text{ mm}$$

6. Standard Thread Selection (Table 11.1)

Select standard coarse thread with $d_c \geq 14.63$ mm

$$d_{\text{c, std}} \geq d_c$$

$$d_c = 14.933 \text{ mm}$$

	$d = 18 \text{ mm}$
7. Design Output	**Select Fastener Designation: M 18 Bolt ($\boldsymbol{d = 18 \text{ mm}, d_c = 14.933 \text{ mm}}$)**

SECTION 10: Boiler Longitudinal Bar Stay Sizing

System Parameters	**Design Protocol**
Pitch of Stays: $350\ \text{mm} \times 350\ \text{mm}$	Calculate area supported by each stay $A = p_s \times p_s$
Steam Pressure: $p = 0.84 \text{N/mm}^2$	Calculate steam force $P = A \cdot p$
Permissible Tensile Stress: $\sigma_t = 56\ \text{MPa} = 56 \text{N/mm}^2$	Solve required core diameter d_c and select bolt size

1. Area Supported by Each Stay (A)

Area of boiler plate supported per stay

$$A = p_s \times p_s$$
$$= 350 \times 350$$
$$= 122500 \text{mm}^2$$

2. Steam Force Acting on Stay (P)

Total steam pressure load on supported area

$$P = A \cdot p$$
$$= 122500 \text{mm}^2 \times 0.84 \text{N/mm}^2$$
$$= 102900\ \text{N}$$

3. Required Core Root Diameter (d_c)

Core diameter required for permissible tension stress

$$P = \frac{\pi}{4}(d_c)^2 \cdot \sigma_t$$

$$102900 = \frac{\pi}{4}(d_c)^2 \times 56$$

$$= 43.98(d_c)^2$$

$$(d_c)^2 = \frac{102900}{43.98}$$

$$= 2339.6$$

$$d_c = \sqrt{2339.6}$$

$$= 48.37 \text{ mm}$$

4. Standard Thread Selection (Table 11.1)

Select standard coarse thread with $d_c \geq 48.37$ mm

$$d_{\text{c, std}} \geq d_c$$

$$d_c = 49.177 \text{ mm}$$

$$d = 56 \text{ mm}$$

5. Design Output

Select Fastener Designation: M 56 Bolt ($d = 56$ mm, $d_c = 49.177$ mm)

SECTION 10: Boiler Bar Stay Arrangement — MECHANICAL DIAGRAM

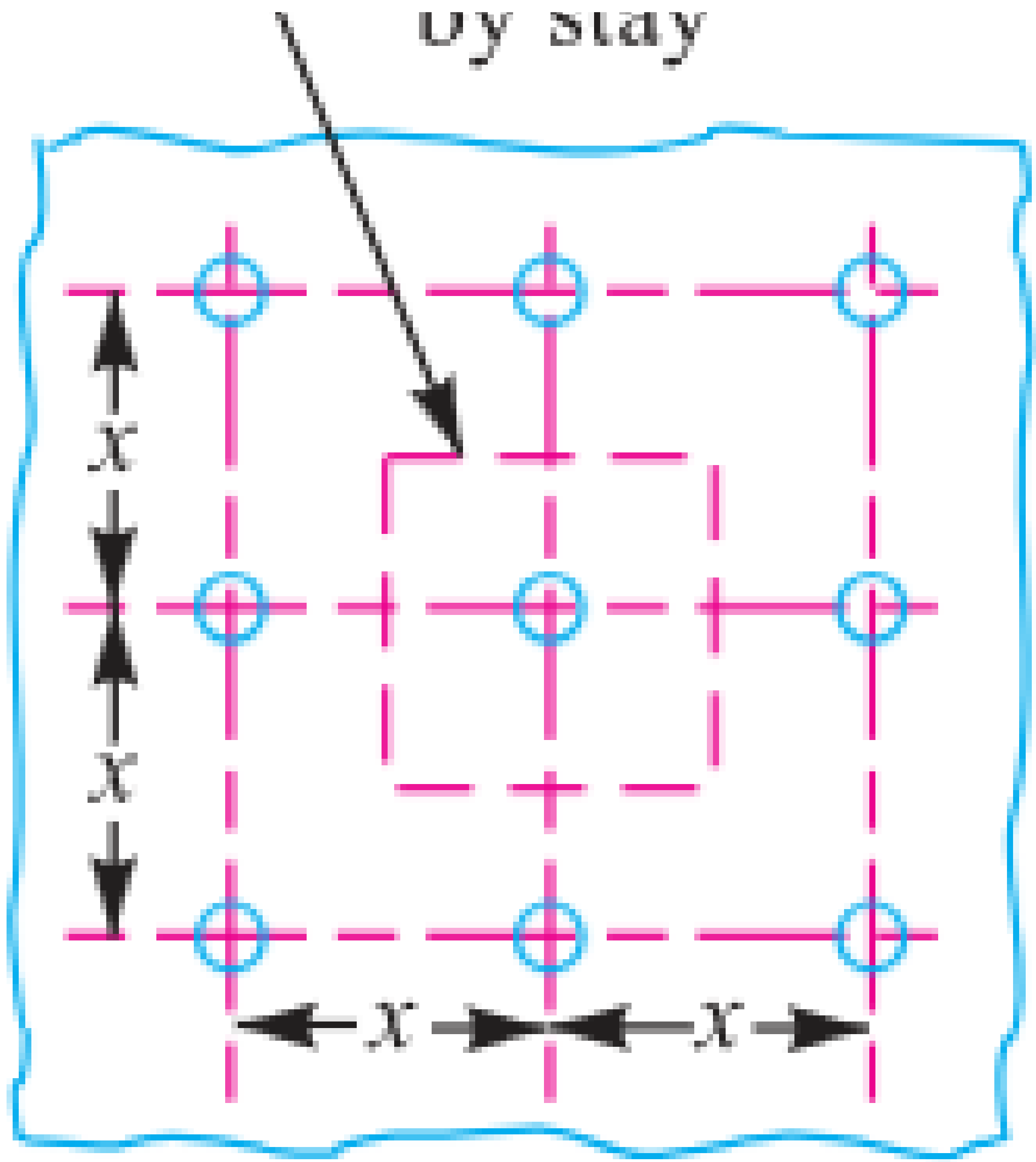


Figure 11.29: Boiler Longitudinal Bar Stay Mounting Scheme

SECTION 11: Turned Shank & Drilled Hole Bolts of Uniform Strength

System Parameters	**Design Protocol**
Nominal Bolt Size: $D_o = 48$ mm (M 48 Coarse Series)	Lookup core root diameter D_c from Table 11.1
Objective: Determine drilled axial hole diameter D for uniform strength	Equate unthreaded drilled shank area to core root area
	Solve $D = \sqrt{D_o^2 - D_c^2}$

1. Core Diameter Lookup (D_c)

Table 11.1 lookup for M 48 coarse thread

$$D_c = f(D_o)$$
$$= 41.795 \text{ mm}$$

2. Area Equality Condition for Uniform Strength

Equate unthreaded drilled shank area to root area

$$\frac{\pi}{4}\left(D_o^2 - D^2\right) = \frac{\pi}{4}D_c^2$$
$$D_o^2 - D^2 = D_c^2$$

3. Drilled Axial Hole Diameter (D)

Solve central axial hole diameter

$$D = \sqrt{D_o^2 - D_c^2}$$

$$= \sqrt{(48)^2 - (41.795)^2}$$

$$= \sqrt{2304 - 1746.82}$$

$$= \sqrt{557.18}$$

$$= \mathbf{23.64\ mm}$$

4. Design Output

Drill Central Axial Hole of Diameter $D = \mathbf{23.64\ mm}$

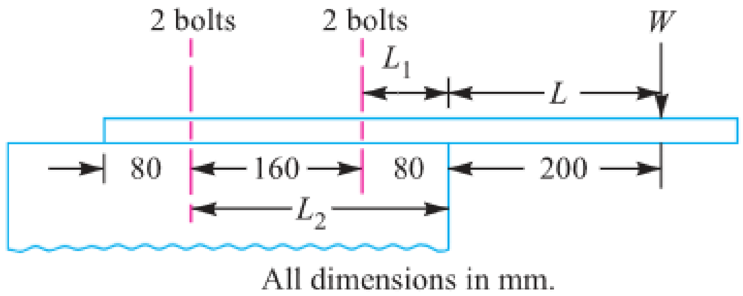


Fig. 11.48

Figure 11.48: Turned Shank Uniform Strength Bolt

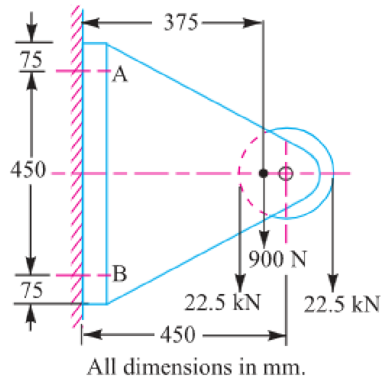


Fig. 11.49

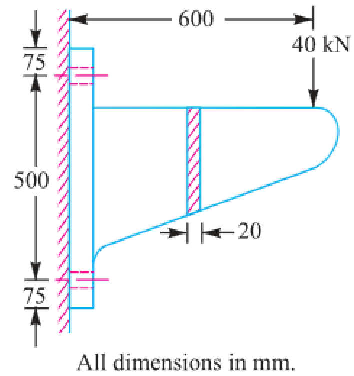


Fig. 11.50

Figure 11.49 & 11.50: Core-Drilled Hole Uniform Strength Bolt

SECTION 12: Wall Brackets under Parallel Eccentric Tensile Loading

System Parameters	**Design Protocol**
Load: $W = 30\ \mathrm{kN} = 30000\ \mathrm{N}$	Calculate direct tensile load $W_{\mathrm{t1}} = \frac{W}{n}$
Eccentricity Arm: $L = 500\ \mathrm{mm}$	Calculate unit secondary load $w = \frac{WL}{2(L_1^2 + L_2^2)}$
Distances: $L_1 = 80\ \mathrm{mm}$, $L_2 = 250\ \mathrm{mm}$	Calculate total tension $W_t = W_{\mathrm{t1}} + wL_2$ and core size d_c
Bolt Count: $n = 4$ (2 per row) | Allowable Stress: $\sigma_t = 60\ \mathrm{MPa}$	

1. Direct Tensile Load per Bolt (W_{t1})

Direct load shared equally by 4 bolts

$$W_{\mathrm{t1}} = \frac{W}{n}$$
$$= \frac{30000}{4}$$
$$= 7500\ \mathrm{N}$$
$$= 7.5\ \mathrm{kN}$$

2. Secondary Tensile Load on Upper Bolts (W_{t2})

Moment equilibrium load factor w

$$w = \frac{W \cdot L}{2\left(L_1^2 + L_2^2\right)}$$

$$= \frac{30 \times 500}{2(80^2 + 250^2)}$$

$$= \frac{15000}{2(6400 + 62500)}$$

$$= 0.10886 \text{ kN/mm}$$

$$W_{t2} = w \cdot L_2$$

$$= 0.10886 \times 250$$

$$= 27.215 \text{ kN}$$

$$= 27215 \text{ N}$$

3. Total Maximum Tensile Load (W_t)

Combined direct and secondary tension on critical top bolts

$$W_t = W_{t1} + W_{t2}$$

$$= 7.5 + 27.215$$

$$= 34.715 \text{ kN}$$

$$= 34715 \text{ N}$$

4. Required Core Diameter (d_c)

Solve root diameter for top row fasteners

$$W_t = \frac{\pi}{4}(d_c)^2 \cdot \sigma_t$$

$$\begin{aligned} 34715 &= \frac{\pi}{4}(d_c)^2 \times 60 \\ &= 47.12(d_c)^2 \\ (d_c)^2 &= \frac{34715}{47.12} \\ &= 736.7 \\ d_c &= \sqrt{736.7} \\ &= 27.14 \text{ mm} \end{aligned}$$

5. Standard Thread Selection (Table 11.1)

Select standard coarse thread with $d_c \geq 27.14 \text{ mm}$

$$d_{\text{c, std}} \geq d_c$$

$$d_c = 28.706 \text{ mm}$$

$$d = 33 \text{ mm}$$

6. Design Output

Select Fastener Designation: M 33 Bolt ($\boldsymbol{d = 33 \text{ mm}, d_c = 28.706 \text{ mm}}$)

11.19 Eccentric Load Acting Parallel to the Axis of Bolts

Consider a bracket having a rectangular base bolted to a wall by means of four bolts as shown in Fig. 11.31. A little consideration will show that each bolt is subjected to a direct tensile load of

$W_{t1} = \frac{W}{n}$, where n is the number of bolts.

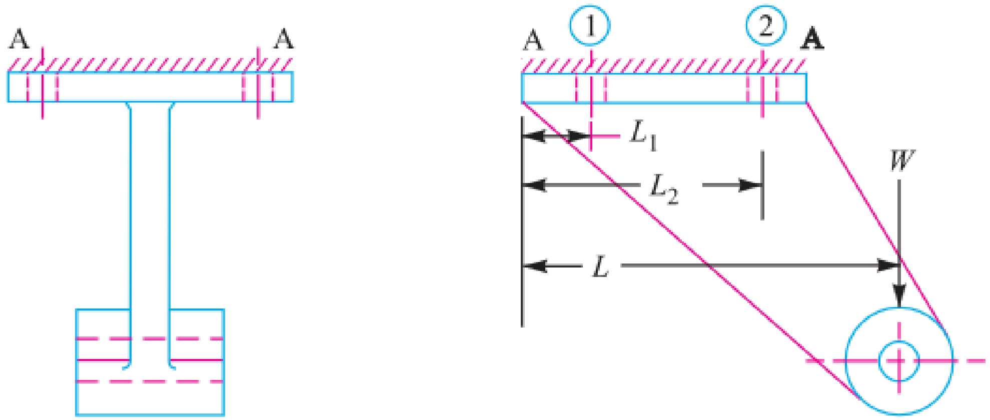


Fig. 11.31. Eccentric load acting parallel to the axis of bolts.

Figure 11.31: Wall Bracket under Parallel Eccentric Load

SECTION 13: Crane Runway T-Bracket Stress Analysis & Fastening Bolts

System Parameters	**Design Protocol**
Wheel Load: $W = 15\ \text{kN} = 15000\ \text{N}$	Part 1: Calculate centroid $\overline{y}$, moment of inertia I_{GG},
Fastening Bolts: $d = 25\ \text{mm}\ (n = 4)$	and max arm stresses σ_t, σ_c
T-Section: Flange 135×25, Web 175×25, Total Depth	Part 2: Calculate direct & secondary bolt tension to
$200\ \text{mm}$	find bolt stress σ_{tb}

1. Centroid ($\overline{y}$) & Section Area (A)

T-section geometry evaluation

$$A_1 = 135 \times 25 = 3375\text{mm}^2$$

$$A_2 = 175 \times 25 = 4375\text{mm}^2$$

$$A = A_1 + A_2$$

$$= 7750\text{mm}^2$$

$$\overline{y} = \frac{A_1 y_1 + A_2 y_2}{A}$$

$$= \frac{3375 \times 12.5 + 4375 \times 112.5}{7750}$$

$$= 68.95 \text{ mm}$$

$$\approx 69 \text{ mm}$$

$$y_1 = 69 \text{ mm} \quad \text{(Top)}$$

$$y_2 = 200 - 69 = 131 \text{ mm} \quad \text{(Bottom)}$$

2. Moment of Inertia (I_{GG}) & Section Moduli (Z_1, Z_2)

Parallel axis theorem for T-section

$$I_{\mathrm{GG}} = \sum (I_i + A_i d_i^2)$$

$$= \left[\frac{135(25)^3}{12} + 3375(56.5)^2\right] + \left[\frac{25(175)^3}{12} + 4375(43.5)^2\right]$$

$$= 30.4 \times 10^6 \text{mm}^4$$

$$Z_1 = \frac{I_{\mathrm{GG}}}{y_1}$$

$$= \frac{30.4 \times 10^6}{69}$$

$$= 440.6 \times 10^3 \mathrm{mm}^3$$

$$Z_2 = \frac{I_{\mathrm{GG}}}{y_2}$$

$$= \frac{30.4 \times 10^6}{131}$$

$$= 232 \times 10^3 \mathrm{mm}^3$$

3. Bracket Arm Stresses at Section X-X

Bending moment $M = 15000(200 + 69) = 4.035 \times 10^6 \ \mathrm{N \cdot mm}$

$$\sigma_{\mathrm{b1}} = \frac{M}{Z_1}$$

$$= \frac{4.035 \times 10^6}{440.6 \times 10^3}$$

$$= 9.16 \mathrm{N/mm}^2$$

$$\sigma_{\mathrm{b2}} = \frac{M}{Z_2}$$

$$= \frac{4.035 \times 10^6}{232 \times 10^3}$$

$$= 17.4 \mathrm{N/mm^2}$$

$$\sigma_{\mathrm{t1}} = \frac{W}{A}$$

$$= \frac{15000}{7750}$$

$$= 1.94 \mathrm{N/mm^2}$$

$$\sigma_t = \sigma_{\mathrm{b1}} + \sigma_{\mathrm{t1}}$$

$$= 9.16 + 1.94$$

$$= \mathbf{11.1\ MPa} \quad (\textbf{Tensile})$$

$$\sigma_c = \sigma_{\mathrm{b2}} - \sigma_{\mathrm{t1}}$$

$$= 17.4 - 1.94$$

$$= \mathbf{15.46\ MPa} \quad (\textbf{Compressive})$$

4. Fastening Bolt Loads (W_{t1}, W_{t2}, W_t)

Tilting edge $\mathbb{E}$: $L_1 = 50$ mm, $L_2 = 375$ mm, $L = 525$ mm

$$W_{t1} = \frac{W}{n}$$

$$= \frac{15000}{4}$$

$$= 3750 \text{ N}$$

$$w = \frac{W \cdot L}{2\left(L_1^2 + L_2^2\right)}$$

$$= \frac{15000 \times 525}{2(50^2 + 375^2)}$$

$$= \frac{7875000}{286250}$$

$$= 27.5 \text{ N/mm}$$

$$W_{t2} = w \cdot L_2$$

$$= 27.5 \times 375$$

$$= 10312.5 \text{ N}$$

$$W_t = W_{t1} + W_{t2}$$
$$= 3750 + 10312.5$$
$$= 14062.5 \text{ N}$$

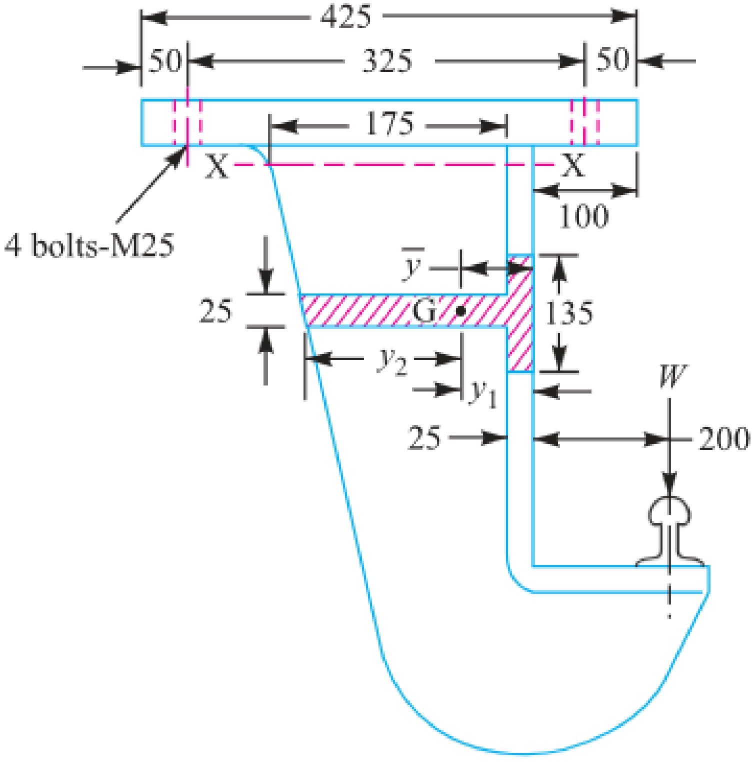
5. Maximum Tensile Stress in Fastening Bolts (σ_{tb})

Bolt core diameter $d_c = 0.84d = 0.84 \times 25 = 21 \text{ mm}$

$$\sigma_{tb} = \frac{W_t}{\frac{\pi}{4}(d_c)^2}$$
$$= \frac{14062.5}{\frac{\pi}{4}(21)^2}$$
$$= \frac{14062.5}{346.36}$$
$$= \mathbf{40.6 \text{ MPa}}$$

6. Design Output

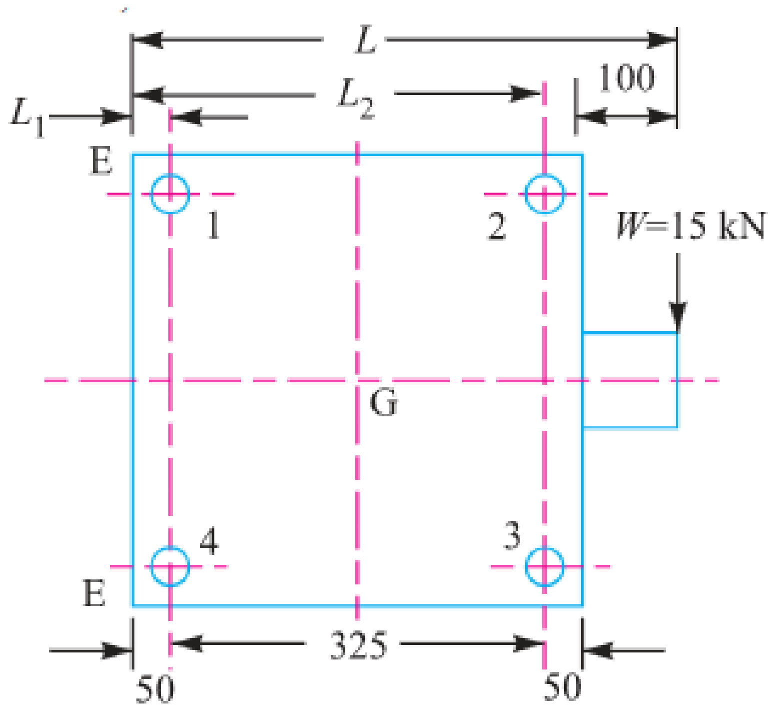
Arm Tensile $\sigma_t = 11.1 \text{ MPa}$, Arm Compressive $\sigma_c = 15.46 \text{ MPa}$, Bolt Stress $\sigma_{tb} = 40.6 \text{ MPa}$



All dimensions in mm

Fig. 11.32

Figure 11.32: Crane Runway T-Bracket Structure



All dimensions in mm.

Fig. 11.33

Figure 11.33: T-Section Stress Distribution Diagram

SECTION 14: Travelling Crane Bracket Sizing (Combined Shear & Tension)

System Parameters	**Design Protocol**
Load: $W = 12 \text{ kN} = 12000 \text{ N}$ Load Arm: $L = 400 \text{ mm}$ Distances: $L_1 = 50 \text{ mm}, L_2 = 375 \text{ mm}$ Permissible Tensile Stress: $\sigma_t = 84 \text{ MPa}$ | Bolt Count: $n = 4$	Part 1: Calculate direct shear W_s, secondary tension W_t, and equivalent load W_{te} via Maximum Principal Stress Theory Part 2: Calculate arm thickness t for depth $b = 250 \text{ mm}$

1. Direct Shear (W_s) & Secondary Tension (W_t)

Primary shear load and secondary tilting tension per bolt

$$
\begin{aligned}
W_s &= \frac{W}{n} \\
&= \frac{12000}{4} \\
&= 3000 \text{ N} \\
&= 3 \text{ kN}
\end{aligned}
$$

$$W_t = \frac{W \cdot L \cdot L_2}{2\left(L_1^2 + L_2^2\right)}$$

$$= \frac{12 \times 400 \times 375}{2(50^2 + 375^2)}$$

$$= \frac{1800000}{286250}$$

$$= 6.288 \text{ kN}$$

$$= 6288 \text{ N}$$

2. Equivalent Tensile Load (W_{te})

Maximum Principal Stress Theory for combined shear & tension

$$W_{\text{te}} = \frac{1}{2}\left[W_t + \sqrt{W_t^2 + 4W_s^2}\right]$$

$$= \frac{1}{2}\left[6.29 + \sqrt{(6.29)^2 + 4(3)^2}\right]$$

$$= \frac{1}{2}[6.29 + 8.69]$$

$$= 7.49 \text{ kN}$$

$$= 7490 \text{ N}$$

3. Required Core Diameter (d_c) & Bolt Selection

Table 11.1 lookup for metric coarse thread

$$W_{\text{te}} = \frac{\pi}{4}(d_c)^2 \cdot \sigma_t$$

$$\begin{aligned} 7490 &= \frac{\pi}{4}(d_c)^2 \times 84 \\ &= 65.97(d_c)^2 \\ (d_c)^2 &= \frac{7490}{65.97} \\ &= 113.5 \\ d_c &= \sqrt{113.5} \\ &= 10.65 \text{ mm} \end{aligned}$$

4. Rectangular Cross-Section of Bracket Arm ($t \times b$)

Bending moment $M = W \cdot L = 12 \times 10^3 \times 400 = 4.8 \times 10^6 \text{ N} \cdot \text{mm}$

$$Z = \frac{1}{6} \cdot t \cdot b^2$$

$$\sigma_t = \frac{M}{Z}$$

$$84 = \frac{4.8 \times 10^6}{\frac{1}{6} \cdot t \cdot b^2}$$

$$t \cdot b^2 = \frac{28.8 \times 10^6}{84}$$

$$= 342857 \text{mm}^3$$

$$t = \frac{342857}{(250)^2}$$

$$= \frac{342857}{62500}$$

$$= \mathbf{5.5\ mm}$$

5. Design Output

Final Design: M 14 Bolts, Bracket Arm Dimensions

$\mathbf{5.5\ mm \times 250\ mm}$

SECTION 14: Travelling Crane Bracket — MECHANICAL DIAGRAM

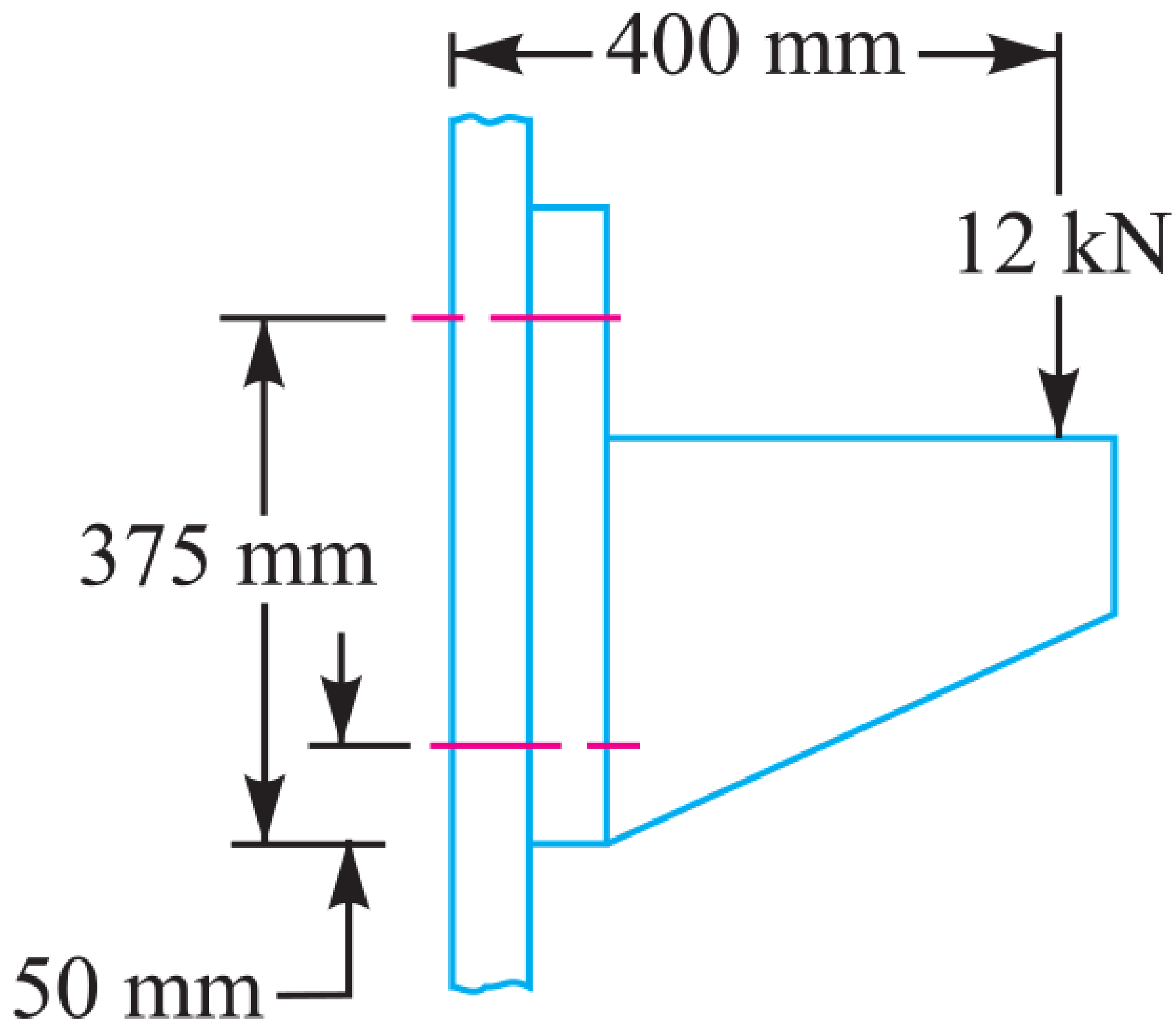


Fig. 11.35

Figure 11.35: Travelling Crane Wall Bracket Structural Geometry

SECTION 15: Inclined Load Bracket Fastener Sizing & Arm Thickness Analysis

System Parameters	**Design Protocol**
Inclined Load: $W = 40 \text{ kN} = 40000 \text{ N}$ at $\theta = 60^{\text{deg}}$ to vertical	Part 1: Resolve load into W_H, W_V and net moment $T_{\text{net}} = T_V - T_H$
Stresses: $\sigma_t = 70 \text{ MPa}, \tau = 50 \text{ MPa}, \sigma_c = 105 \text{ MPa}$	Part 2: Calculate bolt tension W_t, equivalent load W_{te}, and size d_c
Arm Depth: $b = 130 \text{ mm}$ | Bolts: $n = 4, L_1 = 60 \text{ mm}, L_2 = 180 \text{ mm}$	Part 3: Calculate arm thickness t via principal tensile stress $\sigma_{\text{t(max)}}$

1. Force Resolution & Net Turning Moment (T_{net})

Horizontal and vertical force components

$$W_H = W \sin\theta$$
$$= 40 \sin 60^{\text{deg}}$$
$$= 34.64 \text{ kN}$$
$$= 34640 \text{ N}$$

$$W_V = W\cos\theta$$

$$= 40\cos 60^{\text{deg}}$$

$$= 20 \text{ kN}$$

$$= 20000 \text{ N}$$

$$T_H = 34640 \times 20$$

$$= 692.8 \times 10^3 \text{ N} \cdot \text{mm}$$

$$T_V = 20000 \times 175$$

$$= 3500 \times 10^3 \text{ N} \cdot \text{mm}$$

$$T_{\text{net}} = T_V - T_H$$

$$= (3500 - 692.8) \times 10^3$$

$$= 2807.2 \times 10^3 \text{ N} \cdot \text{mm}$$

2. Fixing Bolts Sizing ($W_t, W_s, W_{\text{te}}, d_c$)

Direct & secondary load combination on critical top bolts

$$W_{\text{t1}} = \frac{W_H}{n}$$

$$= 8660 \text{ N}$$

$$W_s = \frac{W_V}{n}$$

$$= 5000 \text{ N}$$

$$w = \frac{T_{\text{net}}}{2\left(L_1^2 + L_2^2\right)}$$

$$= \frac{2807.2 \times 10^3}{2(60^2 + 180^2)}$$

$$= 38.99 \text{ N/mm}$$

$$\approx 39 \text{ N/mm}$$

$$W_{\text{t2}} = w \cdot L_2$$

$$= 39 \times 180$$

$$= 7020 \text{ N}$$

$$W_t = W_{\text{t1}} + W_{\text{t2}}$$

$$= 8660 + 7020$$

$$= 15680 \text{ N}$$

$$W_{\mathrm{te}} = \frac{1}{2}\left[W_t + \sqrt{W_t^2 + 4W_s^2}\right]$$

$$= \frac{1}{2}\left[15680 + \sqrt{(15680)^2 + 4(5000)^2}\right]$$

$$= 17140 \text{ N}$$

$$(d_c)^2 = \frac{17140}{55}$$

$$= 311.64$$

$$d_c = 17.65 \text{ mm}$$

3. Bracket Arm Thickness (t via Principal Stress)

Stresses in upper arm fibre for $b = 130$ mm ($Z = 2817t$)

$$\sigma_{\mathrm{t1}} = \frac{W_H}{A}$$

$$= \frac{34640}{130t}$$

$$= \frac{266.5}{t}$$

$$\sigma_{\text{t2}} = \frac{M_H}{Z}$$

$$= \frac{34640 \times 35}{2817t}$$

$$= \frac{430.4}{t}$$

$$\sigma_{\text{t3}} = \frac{M_V}{Z}$$

$$= \frac{20000 \times 200}{2817t}$$

$$= \frac{1420}{t}$$

$$\sigma_t = \sigma_{\text{t1}} + \sigma_{\text{t2}} + \sigma_{\text{t3}}$$

$$= \frac{266.5 + 430.4 + 1420}{t}$$

$$= \frac{2116.9}{t}$$

$$\tau = \frac{W_V}{A}$$

$$= \frac{20000}{130t}$$

$$= \frac{154}{t}$$

$$\sigma_{\mathrm{t(max)}} = \frac{\sigma_t}{2} + \frac{1}{2}\sqrt{\sigma_t^2 + 4\tau^2}$$

$$= \frac{1058.45}{t} + \frac{1069.6}{t}$$

$$= \frac{2128.05}{t}$$

$$70 = \frac{2128.05}{t}$$

$$t = 30.4 \text{ mm}$$

$$= \mathbf{31 \text{ mm}}$$

4. Design Output

Final Design: M 22 Bolts, Bracket Arm Thickness $t = 31$ mm

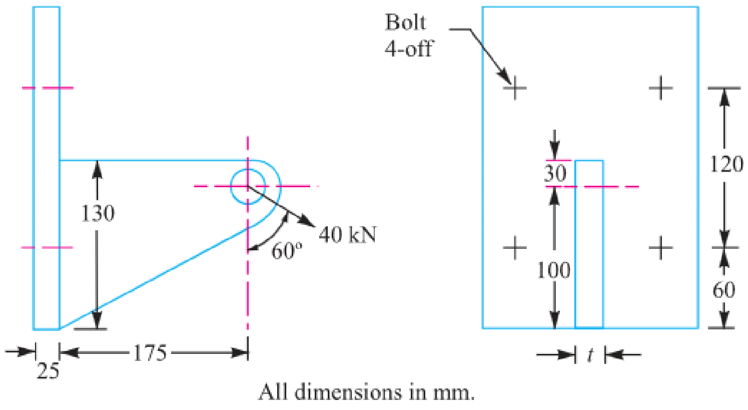


Fig 11.36

Figure 11.36: Wall Bracket under Inclined Load

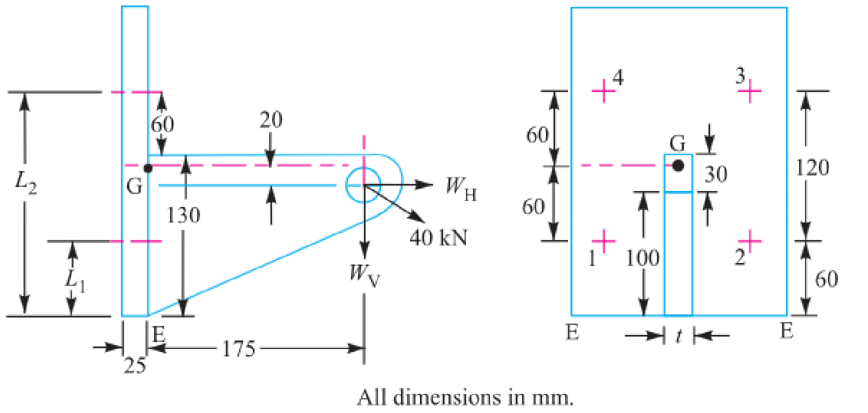


Fig. 11.37

Figure 11.37: Load Resolution & Moment Arm Diagram

SECTION 16: Offset I-Section Bracket under Inclined Loading

System Parameters	**Design Protocol**
Pull Load: $W = 10 \text{ kN} = 10000 \text{ N}$ at $\theta = 60^{\text{deg}}$ to vertical	Part 1: Resolve load into W_H, W_V and net moment $T_{\text{net}} = T_V - T_H$
Stresses: $\sigma_t = 100 \text{ MPa}, \tau = 60 \text{ MPa}$	Part 2: Calculate bolt tension W_t, equivalent load W_{te}, and size d_c
Bolt Count: $n = 4$ | I-Section Flange Width: $b = 3t$	Part 3: Calculate I-section arm thickness t and width $b = 3t$

1. Force Resolution & Net Turning Moment ($\boldsymbol{T_{\text{net}}}$)

Component forces and net moment about tilting edge E

$$W_H = W \sin\theta$$

$$= 10 \sin 60^{\text{deg}}$$

$$= 8.66 \text{ kN}$$

$$= 8660 \text{ N}$$

$$W_V = W\cos\theta$$

$$= 10\cos 60^{\mathrm{deg}}$$

$$= 5\ \mathrm{kN}$$

$$= 5000\ \mathrm{N}$$

$$T_H = 433\ \mathrm{N\cdot m} \quad \text{(Clockwise)}$$

$$T_V = 1500\ \mathrm{N\cdot m} \quad \text{(Counterclockwise)}$$

$$T_{\mathrm{net}} = T_V - T_H$$

$$= 1500 - 433$$

$$= 1067\ \mathrm{N\cdot m}$$

2. Fixing Bolts Sizing ($\boldsymbol{W_t, W_s, W_{\mathrm{te}}, d_c}$)

Bolt distances $L_1 = 0.0375\ \mathrm{m}, L_2 = 0.2125\ \mathrm{m}$

$$W_{\mathrm{t1}} = \frac{W_H}{n}$$

$$= 2165\ \mathrm{N}$$

$$W_s = \frac{W_V}{n}$$

$$= 1250\ \mathrm{N}$$

$$w = \frac{T_{\text{net}}}{2\left(L_1^2 + L_2^2\right)}$$

$$= \frac{1067}{2(0.0375^2 + 0.2125^2)}$$

$$= 11470 \text{ N/m}$$

$$W_{\text{t2}} = w \cdot L_2$$

$$= 11470 \times 0.2125$$

$$= 2435 \text{ N}$$

$$W_t = W_{\text{t1}} + W_{\text{t2}}$$

$$= 2165 + 2435$$

$$= 4600 \text{ N}$$

$$W_{\mathrm{te}} = \frac{1}{2}\left[W_t + \sqrt{W_t^2 + 4W_s^2}\right]$$

$$= \frac{1}{2}\left[4600 + \sqrt{(4600)^2 + 4(1250)^2}\right]$$

$$= 4920 \text{ N}$$

$$(d_c)^2 = \frac{4920}{78.55}$$

$$= 62.63$$

$$d_c = 7.91 \text{ mm}$$

3. Dimensions of I-Section Bracket Arm (t and $b = 3t$)

Cross-section area $A = 9t^2$, section modulus $Z = 10.7t^3$

$$\sigma_{\mathrm{t1}} = \frac{W_H}{A}$$

$$= \frac{8660}{9t^2}$$

$$= \frac{962}{t^2}$$

$$\sigma_{\mathrm{t2}} = \frac{M_H}{Z}$$

$$= \frac{433 \times 10^3}{10.7t^3}$$

$$= \frac{40.5 \times 10^3}{t^3}$$

$$\sigma_{\mathrm{t3}} = \frac{M_V}{Z}$$

$$= \frac{1500 \times 10^3}{10.7t^3}$$

$$= \frac{140.2 \times 10^3}{t^3}$$

$$\sigma_t = \sigma_{\mathrm{t1}} - \sigma_{\mathrm{t2}} + \sigma_{\mathrm{t3}}$$

$$= \frac{962}{t^2} + \frac{99.7 \times 10^3}{t^3}$$

$$100 = \frac{962}{t^2} + \frac{99.7 \times 10^3}{t^3} \Rightarrow \boldsymbol{t = 10.4\ \mathrm{mm}}$$

$$\boldsymbol{b = 3t = 31.2\ \mathrm{mm}}$$

4. Design Output	**Final Design: M 10 Bolts, I-Section Arm Dimensions $t = 10.4\ \text{mm}, b = 31.2\ \text{mm}$**

SECTION 17: Pillar Crane Circular Base Foundation Flange Sizing

System Parameters	Design Protocol
Bolt Count: $n = 8$	Calculate distance from tilting edge $A - A$: $L = e - R$
Bolt Circle Diameter: $d = 1.6\ \mathrm{m} \Rightarrow r = 0.8\ \mathrm{m}$	Calculate maximum tensile load W_t on critical bolt using PCD formula
Pillar Base Diameter: $D = 2\ \mathrm{m} \Rightarrow R = 1\ \mathrm{m}$	Solve required core root diameter d_c and select bolt size
Crane Load: $W = 100\ \mathrm{kN} = 100000\ \mathrm{N}$ at distance $e = 5\ \mathrm{m}$ | Allowable Stress: $\sigma_t = 100\ \mathrm{MPa}$	

1. Tilting Arm Distance (L)

Distance measured from base fulcrum edge $A - A$ of radius $R = 1\ \mathrm{m}$

$$
\begin{aligned}
L &= e - R \\
&= 5 - 1 \\
&= 4\ \mathrm{m}
\end{aligned}
$$

2. Maximum Tensile Load on Critical Bolt (W_t)

Circular foundation bolt layout equation for tilting about base edge

$$W_t = \frac{2 \cdot W \cdot L \cdot (R + r)}{n(2R^2 + r^2)}$$

$$= \frac{2 \times (100000) \times 4 \times (1 + 0.8)}{8 \times (2(1)^2 + 0.8^2)}$$

$$= \frac{800000 \times 1.8}{8 \times 2.64}$$

$$= \frac{1440000}{21.12}$$

$$= 68.18 \times 10^3 \text{ N}$$

$$= 68180 \text{ N}$$

3. Required Core Root Diameter (d_c)

Solve root diameter for allowable stress

$$W_t = \frac{\pi}{4}(d_c)^2 \cdot \sigma_t$$

$$\begin{aligned} 68180 &= \frac{\pi}{4}(d_c)^2 \times 100 \\ &= 78.54(d_c)^2 \\ (d_c)^2 &= \frac{68180}{78.54} \\ &= 868.1 \\ d_c &= \sqrt{868.1} \\ &= 29.46 \text{ mm} \end{aligned}$$

4. Standard Thread Selection (Table 11.1)

Select standard coarse thread with $d_c \geq 29.46$ mm

$$d_{\text{c, std}} \geq d_c$$

$$d_c = 31.093 \text{ mm}$$

$$d = 36 \text{ mm}$$

5. Design Output

Select Fastener Designation: M 36 Bolt ($d = 36$ mm, $d_c = 31.093$ mm)

SECTION 17: Pillar Crane Foundation Flange — MECHANICAL DIAGRAM

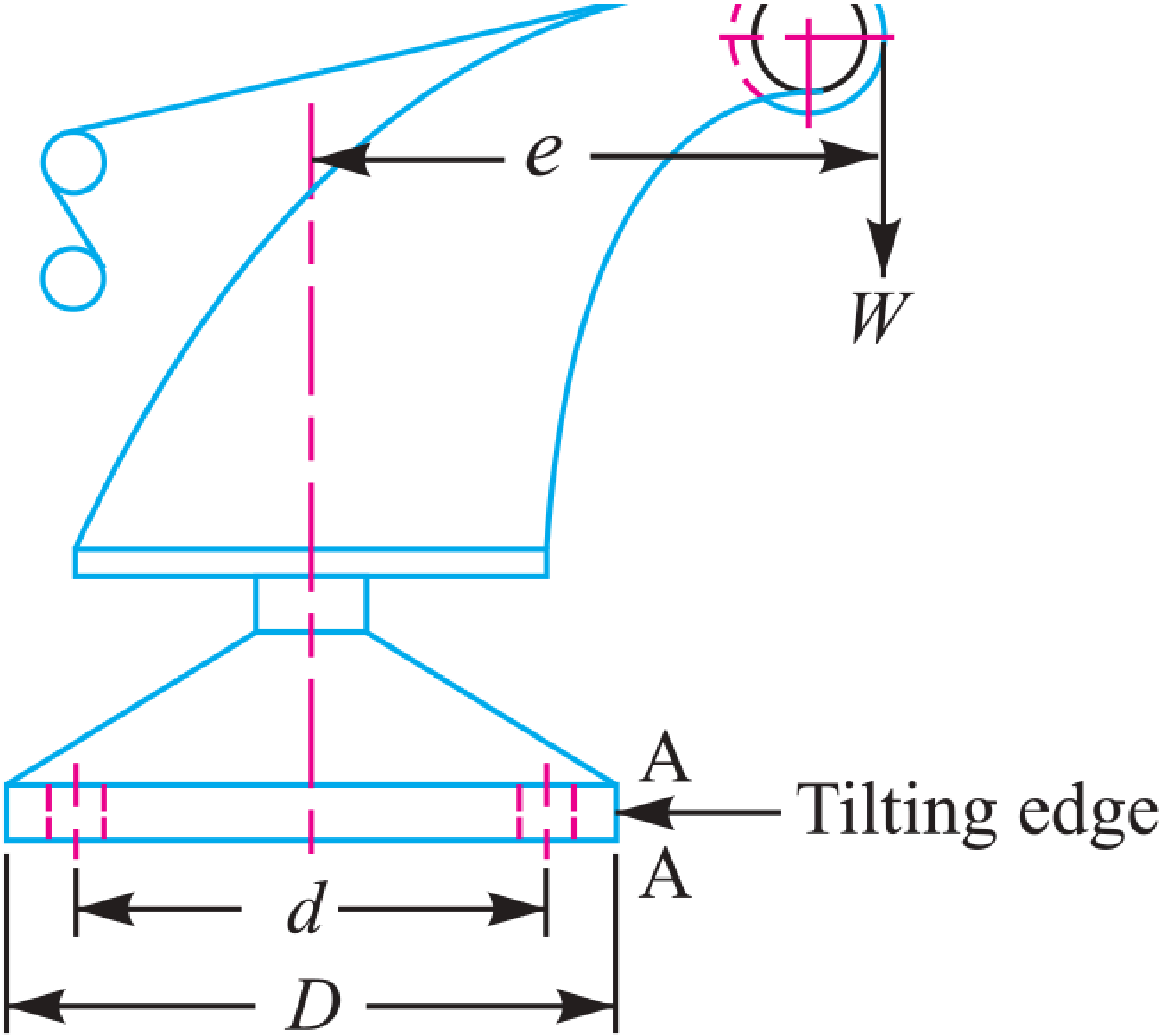


Figure 11.42: Pillar Crane Base Flange 8-Bolt PCD Layout

SECTION 18: Circular Flanged Bearing Base Fastener Sizing

System Parameters	**Design Protocol**
Bolt Count: $n = 4$	For $n = 4$ bolts on circular flange, critical bolt lies at angle $\alpha = \frac{180^{\text{deg}}}{n} = 45^{\text{deg}}$
Bolt Circle Diameter: $d = 500 \text{ mm} \Rightarrow r = 250 \text{ mm}$	Apply circular PCD overturning equation for W_t
Bearing Flange Diameter: $D = 650 \text{ mm} \Rightarrow R = 325 \text{ mm}$	Solve required core root diameter d_c
Overturning Load: $W = 400 \text{ kN} = 400000 \text{ N}$ at arm $L = 250 \text{ mm}$ | Stress: $\sigma_t = 60 \text{ MPa}$	

1. Maximum Tensile Load on Critical Bolt (W_t)

Overturning equation for $n = 4$ bolts at $\alpha = 45^{\text{deg}}$

$$W_t = \frac{2 \cdot W \cdot L \cdot \left[R + r\cos\left(\frac{180^{\text{deg}}}{n}\right)\right]}{n(2R^2 + r^2)}$$

$$= \frac{2 \times (400000) \times 250 \times \left[325 + 250\cos 45^{\text{deg}}\right]}{4 \times (2(325)^2 + (250)^2)}$$

$$= \frac{200 \times 10^6 \times [325 + 176.78]}{4 \times 273750}$$

$$= \frac{200 \times 10^6 \times 501.78}{1095000}$$

$$= 91643 \text{ N}$$

2. Required Core Root Diameter (d_c)

Solve root diameter for allowable stress

$$W_t = \frac{\pi}{4}(d_c)^2 \cdot \sigma_t$$

$$91643 = \frac{\pi}{4}(d_c)^2 \times 60$$

$$= 47.13(d_c)^2$$

$$(d_c)^2 = \frac{91643}{47.13}$$

$$= 1944.5$$

$$d_c = \sqrt{1944.5}$$

$$= 44.1 \text{ mm}$$

3. Standard Thread Selection (Table 11.1)

Select standard coarse thread with $d_c \geq 44.1 \text{ mm}$

$$d_{\text{c, std}} \geq d_c$$

$$d_c = 45.795 \text{ mm}$$

$$d = 52 \text{ mm}$$

4. Design Output

Select Fastener Designation: M 52 Bolt ($d = 52 \text{ mm}, d_c = 45.795 \text{ mm}$)

11.21 Eccentric Load on a Bracket with Circular Base

Sometimes the base of a bracket is made circular as in case of a flanged bearing of a heavy machine tool and pillar crane etc. Consider a round flange bearing of a machine tool having four bolts as shown in Fig. 11.40.

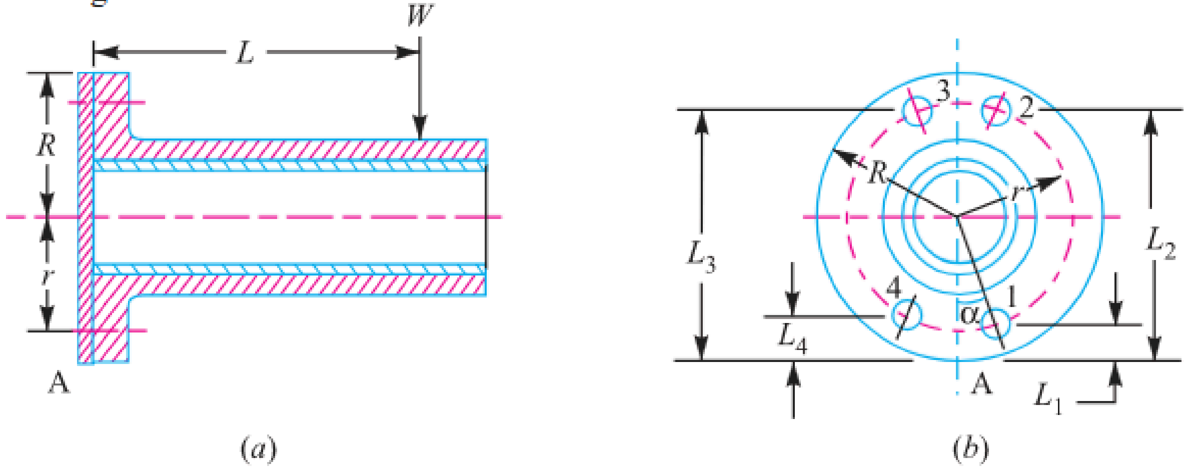


Fig. 11.40. Eccentric load on a bracket with circular base.

Figure 11.40: Circular Flanged Bearing Base PCD Layout

SECTION 19: Pillar Crane Circular Foundation 2-Line Overturning Analysis

System Parameters

Base Diameter: $D = 600\ \mathrm{mm} \Rightarrow R = 300\ \mathrm{mm} = 0.3\ \mathrm{m}$

PCD: $d = 500\ \mathrm{mm} \Rightarrow r = 250\ \mathrm{mm} = 0.25\ \mathrm{m}$

Fasteners: $n = 4$ M 30 bolts ($A_c = 561\mathrm{mm}^2$)

Overturning Load: $W = 60\ \mathrm{kN} = 60000\ \mathrm{N}$ | Allowable Stress: $\sigma_t = 60\ \mathrm{MPa}$

Design Protocol

Part 1: Analyze line X-X (45^{deg} to bolts) to find max distance e

Part 2: Analyze line Y-Y (in line with bolts) at same e to find max induced stress $\sigma_{\max}$

1. Line X-X Analysis (45^{deg} Tilting)

Tensile capacity $P_{\mathrm{cap}} = 561 \times 60 = 33.66\ \mathrm{kN}$, direct load $W_{\mathrm{t1}} = \frac{60}{4} = 15\ \mathrm{kN}$

$$\begin{aligned} P_{\max} &= P_{\mathrm{cap}} + W_{\mathrm{t1}} \\ &= 33.66 + 15 \\ &= 48.66\ \mathrm{kN} \end{aligned}$$

$$\begin{aligned} L_1 &= R - r\cos 45^{\mathrm{deg}} \\ &= 0.3 - 0.25(0.7071) \\ &= 0.123\ \mathrm{m} \end{aligned}$$

$$L_2 = R + r\cos 45^{\mathrm{deg}}$$

$$= 0.477\ \mathrm{m}$$

$$w = \frac{P_{\mathrm{max}}}{L_2}$$

$$= \frac{48.66}{0.477}$$

$$= 102\ \mathrm{kN/m}$$

$$M_{\mathrm{res}} = 2 \cdot w \cdot \left[L_1^2 + L_2^2\right]$$

$$= 2 \times 102 \times \left[(0.123)^2 + (0.477)^2\right]$$

$$= 49.4\ \mathrm{kN \cdot m}$$

2. Maximum Load Distance (e)

Overturning moment $M_{\text{overturn}} = W(e - R) = 60(e - 0.3)$

$$M_{\text{overturn}} = W(e - R)$$

$$\begin{aligned} 60(e - 0.3) &= 49.4 \\ e - 0.3 &= \frac{49.4}{60} \\ &= 0.8233 \text{ m} \\ e &= 0.8233 + 0.3 \\ &= \mathbf{1.123 \text{ m}} \end{aligned}$$

3. Line Y-Y Analysis (In-Line with Bolts)

Distances: $L_1 = 0.05 \text{ m}, L_2 = 0.3 \text{ m}, L_3 = 0.55 \text{ m}$

$$\begin{aligned} M_{\text{res}} &= w \cdot \left[L_1^2 + 2L_2^2 + L_3^2\right] \\ &= w \cdot \left[(0.05)^2 + 2(0.3)^2 + (0.55)^2\right] \\ &= 0.485 \cdot w \\ 0.485 \cdot w &= 49.4 \\ w &= \frac{49.4}{0.485} \\ &= 102 \text{ kN/m} \end{aligned}$$

4. Line Y-Y Maximum Induced Stress ($\sigma_{\max}$)

Secondary load $W_{t2} = w \cdot L_3 = 102 \times 0.55 = 56.1$ kN

$$P_{\text{net}} = W_{t2} - W_{t1}$$
$$= 56.1 - 15$$
$$= 41.1 \text{ kN}$$
$$= 41100 \text{ N}$$

$$\sigma_{\max} = \frac{P_{\text{net}}}{A_c}$$
$$= \frac{41100}{516}$$
$$= \mathbf{79.65\ MPa}$$

5. Design Output

Final Answers: (1) Maximum Distance $e = 1.123$ m, (2) Maximum Induced Stress $\sigma_{\max} = 79.65$ MPa

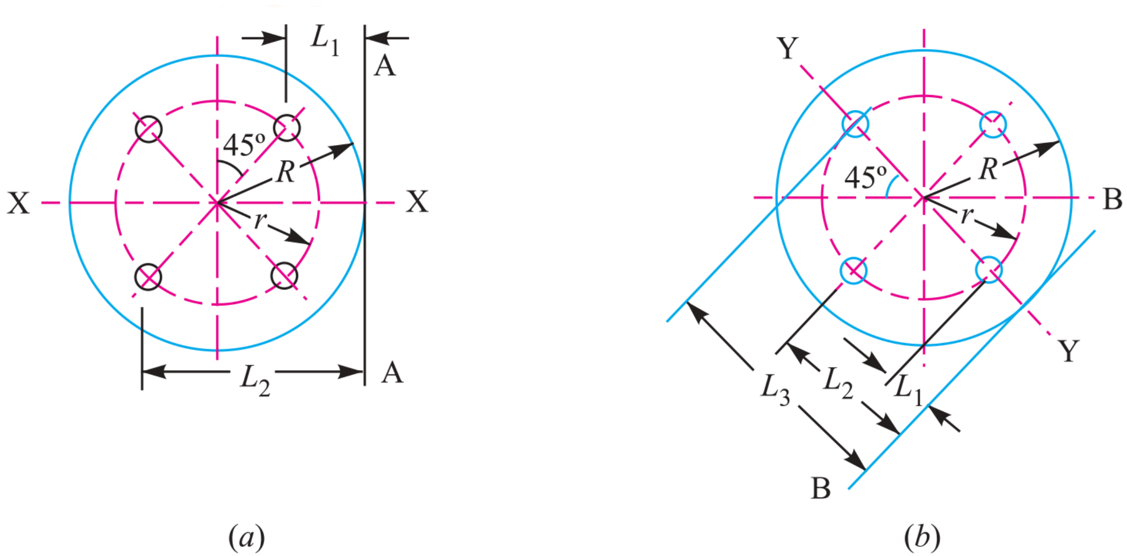


Fig. 11.43

Figure 11.43: 2-Line Overturning Axis Analysis (X-X vs Y-Y)

SECTION 20: Solid Forged Bracket under Combined Bending, Torsion & Coplanar Shear

System Parameters	Design Protocol
Vertical Load: $W = 13.5 \text{ kN} = 13500 \text{ N}$	Part 1: Size arm dia D for combined bending (M) and torsion (T)
Eccentricity: $e = 250 \text{ mm}$	Part 2: Size arm dia d for pure bending (M)
Stresses: Tensile $\sigma_t = 110 \text{ MPa}$, Shear $\tau = 65 \text{ MPa}$ |	Part 3: Calculate top bolt tensile load W_t
Flange: 4 bolts in square layout	Part 4: Calculate maximum resultant shear force W_s on bolts

1. Diameter D for Arm (Combined Bending + Torsion)

$M = 13500 \times (300 - 25) = 3.7125 \times 10^6 \text{ N} \cdot \text{mm}$

$T = 13500 \times 250 = 3.375 \times 10^6 \text{ N} \cdot \text{mm}$

$$T_e = \sqrt{M^2 + T^2}$$

$$= \sqrt{(3.7125 \times 10^6)^2 + (3.375 \times 10^6)^2}$$

$$= 5.017 \times 10^6 \text{ N} \cdot \text{mm}$$

$$T_e = \frac{\pi}{16} D^3 \cdot \tau$$

$$5.017 \times 10^6 = \frac{\pi}{16} D^3 \times 65$$

$$= 12.76 D^3$$

$$D^3 = \frac{5.017 \times 10^6}{12.76}$$

$$= 393.2 \times 10^3$$

$$D = \sqrt[3]{393.2 \times 10^3}$$

$$= 73.24 \text{ mm}$$

$$= \mathbf{75\ mm}$$

2. Diameter d for Arm (Pure Bending)

Bending moment $M = 13500 \times (250 - 37.5) = 2.8688 \times 10^6 \text{ N} \cdot \text{mm}$

$$\sigma_t = \frac{M}{Z}$$

$$110 = \frac{2.8688 \times 10^6}{\frac{\pi}{32} d^3}$$

$$= \frac{2.8688 \times 10^6}{0.0982 d^3}$$

$$d^3 = \frac{29.214 \times 10^6}{110}$$

$$= 265.58 \times 10^3$$

$$d = \sqrt[3]{265.58 \times 10^3}$$

$$= 64.28 \text{ mm}$$

$$= \mathbf{65\ mm}$$

3. Tensile Load on Each Top Bolt (W_t)

Tilting distances $L_1 = 37.5 \text{ mm}, L_2 = 237.5 \text{ mm}$

$$M_{\text{res}} = 2 \cdot w \cdot \left(L_1^2 + L_2^2\right)$$

$$= 2 \cdot w \cdot (37.5^2 + 237.5^2)$$

$$= 115625 \cdot w$$

$$M_{\text{overturn}} = W \cdot L$$
$$= 13500 \times 300$$
$$= 4.05 \times 10^6 \text{ N} \cdot \text{mm}$$
$$w = \frac{4.05 \times 10^6}{115625}$$
$$= 35.03 \text{ N/mm}$$

$$W_t = w \cdot L_2$$
$$= 35.03 \times 237.5$$
$$= 8320 \text{ N}$$
$$= \mathbf{8.32\ kN}$$

4. Maximum Shearing Force on Each Bolt (W_s)

Primary shear $W_{\text{s1}} = \frac{13500}{4} = 3375$ N, radial dist $r_i = \sqrt{100^2 + 100^2} = 141.4$ mm

$$W_{\text{s2}} = \frac{W \cdot e \cdot r_i}{\sum r_i^2}$$
$$= \frac{13500 \times 250 \times 141.4}{4 \times (141.4)^2}$$
$$= 5967 \text{ N}$$

$$W_s = \sqrt{W_{s1}^2 + W_{s2}^2 - 2W_{s1}W_{s2}\cos\theta}$$

$W_s \quad \text{(Bolts 1 \& 4)} = 4303 \text{ N}$

$$W_s = \sqrt{W_{s1}^2 + W_{s2}^2 + 2W_{s1}W_{s2}\cos\theta}$$

$W_s \quad \text{(Bolts 2 \& 3)} = \mathbf{8687\ N}$

5. Design Output

Final Answers: (1) $D = 75$ mm, (2) $d = 65$ mm, (3) Top Bolt $W_t = 8.32$ kN, (4) Max Shear Force $W_s = 8687$ N

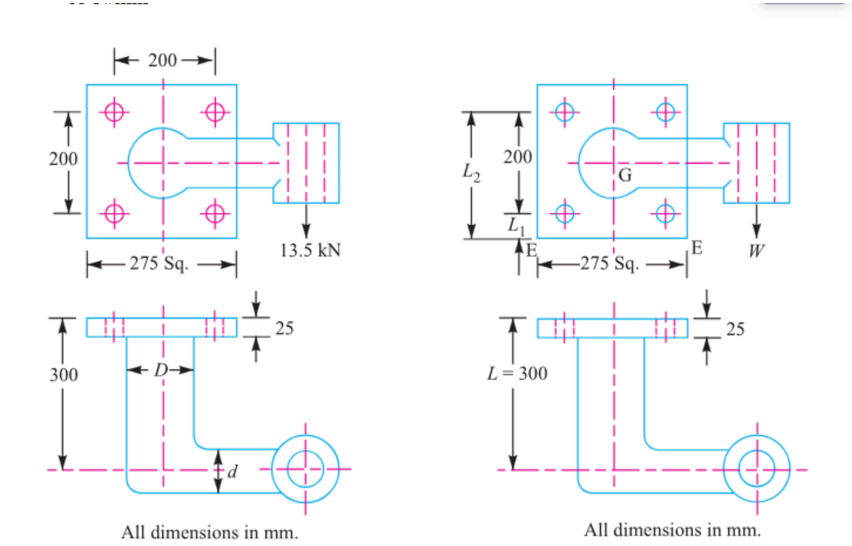


Fig. 11.45 Fig. 11.46

Figure 11.45 & 11.46: Solid Forged Bracket Structure

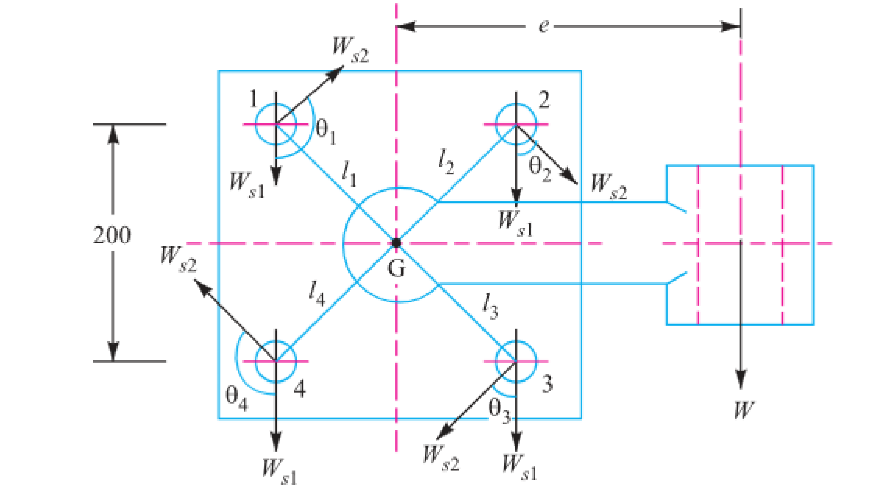


Fig. 11.47

Figure 11.47: Vector Shear Load Diagram on Fastener Group